WORKED SOLUTION

STEP II 2018, Question 10

Pure • Ant on elastic string

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Read the question before you go on. It is in the STEP Support Programme's question database:

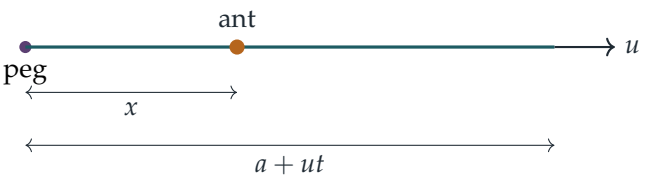
step.maths.org/questions/18-s2-q10

This question has a full guided treatment in *Beyond STEP*, Section 11.12, *The Ant on the Rubber Rope*: https://beyondstep.keithmcnulty.org/ch11/rubber-rope.html

The key idea

Because the string is uniform, it stretches by the same factor everywhere, so each point of the string keeps its *fraction* of the way along. That makes the speed of a point proportional to its distance from the peg. The ant's equation is then linear, and the natural quantity to follow is the ant's fraction $x/(a+ut)$ of the way along the string, rather than x itself.

Solution



Speed of a point of the string

At time t the free end is a distance $a+ut$ from the peg, so that is the length of the string. The string is uniform, so it is stretched by the same factor $(a+ut)/a$ along its whole length. A point of the string that was a distance x_0 from the peg at $t=0$ is therefore always the same fraction x_0/a of the way along:

$$x = \frac{x_0}{a}(a+ut).$$

Differentiating with x_0 fixed, its speed is $\dfrac{x_0 u}{a}$, and writing this in terms of its current position x,

$$\text{speed of the point at distance } x = \frac{ux}{a+ut}.$$

As a check, this is 0 at the peg and u at the free end.

The ant's equation

The ant's speed relative to the peg is v plus the speed of the string beneath it, so

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v + \frac{ux}{a+ut}.$$

By the quotient rule, and then substituting this equation,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) &= \frac{1}{a+ut}\frac{\mathrm{d}x}{\mathrm{d}t} - \frac{ux}{(a+ut)^2} \\ &= \frac{1}{a+ut}\left(v + \frac{ux}{a+ut}\right) - \frac{ux}{(a+ut)^2} = \frac{v}{a+ut},\end{aligned}$$

as required.

The time to reach the end

Integrate from 0 to t, using $x = 0$ at $t = 0$:

$$\frac{x}{a+ut} = \int_0^t \frac{v}{a+us}\,\mathrm{d}s = \frac{v}{u}\ln\left(\frac{a+ut}{a}\right) = \frac{1}{k}\ln\left(1+\frac{ut}{a}\right).$$

The left-hand side is the fraction of the string the ant has covered, and the ant reaches the end when it equals 1. The right-hand side increases steadily from 0 without bound, so this happens exactly once, at the time T with

$$\ln\left(1+\frac{uT}{a}\right) = k \iff 1+\frac{uT}{a} = \mathrm{e}^k \iff uT = a(\mathrm{e}^k - 1),$$

which is the given result.

The journey back

Now the ant walks towards the peg, so its own speed along the string is $-v$, and

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -v + \frac{ux}{a+ut}.$$

Ask yourself Do we need to redo the quotient-rule calculation?

No: the working above is unchanged apart from the sign of v, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{x}{a+ut}\right) = -\frac{v}{a+ut}.$$

The journey back starts at $t = T$ with the ant at the end of the string, where $x/(a+ut) = 1$. Integrating from T to t,

$$\frac{x}{a+ut} = 1 - \frac{1}{k}\ln\left(\frac{a+ut}{a+uT}\right).$$

The ant reaches the peg when this fraction is 0, which first happens at the time t_1 with

$$\ln\left(\frac{a+ut_1}{a+uT}\right) = k \iff a+ut_1 = (a+uT)\,\mathrm{e}^k.$$

So the journey back takes

$$t_1 - T = \frac{(a+uT)(\mathrm{e}^k-1)}{u}.$$

From the outward journey, $\mathrm{e}^k - 1 = uT/a$ and $a+uT = a\mathrm{e}^k$, so

$$t_1 - T = \frac{a\mathrm{e}^k}{u}\cdot\frac{uT}{a} = T\mathrm{e}^k.$$

The journey back takes $T\mathrm{e}^k$, longer than the outward trip by the factor e^k.

Checks

- If $u \to 0$ the string does not stretch: $k \to 0$ and $T = a(\mathrm{e}^k-1)/u \to a/v$, while the return time $T\mathrm{e}^k \to a/v$ too, as it should.
- The return time is longer, which makes sense: on the way back the string beneath the ant carries it away from the peg.
- Stepping both differential equations forward numerically for random values of a, u and v (for example $a = 1$, $u = v$, so $k = 1$ and $T = \mathrm{e} - 1 \approx 1.718$) puts the ant at the end at time T and back at the peg after a further $T\mathrm{e} \approx 4.67$, matching the formulas.

Common slips

- Taking every point of the string to move at speed u; only the free end does.
- Forgetting that the return journey starts at $t = T$, not $t = 0$, so the constant of integration changes.
- Leaving the answer as $(a+uT)(\mathrm{e}^k-1)/u$ when it is asked for in terms of T and k alone.

Beyond STEP

The ant always reaches the end, however fast the string stretches, because $\int \mathrm{d}t/(a+ut)$ diverges. Its distance covered as a fraction of the string grows like the logarithm of the

time, and if the stretching were faster (say exponential) the integral would converge and the ant might never arrive. The same idea, of a fixed speed through space that is itself expanding, appears in cosmology, where it decides whether light from a distant galaxy can ever reach us. The linear equation solved here by spotting the right quantity is a case of the integrating-factor method; see Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.